Feng-Chun Hsu

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RESEARCH PROFILE

Researcher in biomedical optics and computational imaging, combining optical-system development, physical modeling, and physics-informed learning for fast, high-dimensional biological imaging.

EDUCATION

Ph.D. in Photonics Sep 2018 – May 2025
National Yang Ming Chiao Tung University | Tainan, Taiwan | Advisor: Prof. Shean-Jen Chen

M.S. in Engineering Science Sep 2016 – Jul 2018
National Cheng Kung University | Tainan, Taiwan

B.S. in Engineering Science Sep 2012 – Jul 2016
National Cheng Kung University | Tainan, Taiwan

RESEARCH EXPERIENCE

Postdoctoral Research Associate Jul 2026 – Present
Adaptive Photonics Lab, National Yang Ming Chiao Tung University | Tainan, Taiwan | Supervisor: Prof. Shean-Jen Chen

- Develop physics-informed reconstruction methods for light-field microscopy and volumetric biological imaging.
- Model spatiotemporal light propagation in temporal-focusing systems for multiphoton imaging and 3D optical stimulation.

Visiting Scholar Sep 2023 – Sep 2024
So Lab, Department of Mechanical Engineering, Massachusetts Institute of Technology | Advisor: Prof. Peter T. C. So

High-Speed Multiphoton Imaging Through Scattering Media

- Built and calibrated a DMD-based line-scan temporal-focusing system for high-speed multiphoton imaging through scattering media.
- Developed vectorial simulations of line-scan temporal focusing and the associated imaging path to guide system design and alignment.
- Designed and assembled a high-speed detection path using a silicon photomultiplier (SiPM).

Doctoral Research Assistant 2018 – 2025
College of Photonics, National Yang Ming Chiao Tung University | Tainan, Taiwan | Advisor: Prof. Shean-Jen Chen

Nonlinear and Computational Microscopy

- Designed and built temporal-focusing multiphoton systems for bioimaging, including a scanless light-field microscope for volumetric acquisition and reconstruction.
- Developed SLM-based computer-generated holography for spatially confined 3D multiphoton stimulation with temporal focusing.
- Co-developed a four-channel FPGA-controlled DAC and hybrid photodetector readout for high-dynamic-range laser-scanning fluorescence microscopy.

Adaptive Optics and Wavefront Sensing

- Built adaptive-optics systems using SLM and deformable-mirror correction with Shack-Hartmann wavefront sensing.

- Developed and trained neural-network models to reconstruct wavefront and beam intensity from Shack-Hartmann sensor patterns.
- Integrated deformable-mirror and SLM control through ROS and co-developed a compact wavefront sensor for laser micromachining in an academic-industry collaboration with Contrel Technology.

Laser and Imaging Integration

- Developed an integrated laser and vision system for rapid pest control on an unmanned vehicle.

ADDITIONAL EXPERIENCE

Military Service | Taiwan — Jul 2025 – Jul 2026

TECHNICAL EXPERTISE

Imaging: Multiphoton microscopy; temporal focusing; light-field and volumetric microscopy; fluorescence and second-harmonic-generation imaging

Computational: Physical and vectorial modeling; image reconstruction; inverse problems; deep learning; physics-informed inference

Optical & electronic systems: DMD and SLM beam shaping; Shack-Hartmann sensing; adaptive optics; SiPM and hybrid photodetector readout; FPGA/DAC control; ROS integration

PEER-REVIEWED JOURNAL ARTICLES

- **Feng-Chun Hsu**, et al. “Richardson-Lucy-Based Deconvolution Network for Light Field Microscopy.” *Manuscript in preparation.*
- **Feng-Chun Hsu**, Cheng Zheng, Shean-Jen Chen and Peter T.C. So. “Vectorial spatiotemporal modeling of line scanning temporal focusing.” *Manuscript in preparation.*
- **Feng-Chun Hsu**, Chun-Yu Lin, Chia-Yuan Chang, and Shean-Jen Chen. "Microlens Array-Based Beam Profile and Wavefront Sensor With Physical Constraint Learning." *IEEE Photonics Journal* 17, no. 3 (2025): 1-6. DOI
- Liang-Wei Chen, Shang-Yang Lu, **Feng-Chun Hsu**, Chun-Yu Lin, Ann-Shyn Chiang, and Shean-Jen Chen. "Deep-Computer-Generated Holography With Temporal Focusing and a Digital Propagation Matrix for Rapid 3D Multiphoton Stimulation." *Optics Express* 32, no. 2 (2024): 2321-2332. DOI
- Anupama Nair, Chun-Yu Lin, **Feng-Chun Hsu**, Ta-Hsiang Wong, Shu-Chun Chuang, Yi-Shan Lin, Chung-Hwan Chen, Paul Campagnola, Chi-Hsiang Lien, and Shean-Jen Chen. "Categorization of Collagen Type I and II Blend Hydrogel Using Multipolarization SHG Imaging With ResNet Regression." *Scientific Reports* 13 (2023): 19534. DOI
- **Feng-Chun Hsu**, Chun-Yu Lin, Yvonne Yuling Hu, Yeu-Kuang Hwu, Ann-Shyn Chiang, and Shean-Jen Chen. "Light-Field Microscopy With Temporal Focusing Multiphoton Illumination for Scanless Volumetric Bioimaging." *Biomedical Optics Express* 13, no. 12 (2022): 6610-6620. DOI
- Chia-Yuan Chang, Chun-Yun Lin, Yvonne Y. Hu, Sheng-Feng Tsai, **Feng-Chun Hsu**, and Shean-Jen Chen. "Temporal Focusing Multiphoton Microscopy With Optimized Parallel Multiline Scanning for Fast Biotissue Imaging." *Journal of Biomedical Optics* 26, no. 1 (2021): 016501. DOI

CONFERENCE PROCEEDINGS AND PRESENTATIONS

- Jian Zhao, Udith Haputhanthri, Mithunjha Anandakumar, Cheng Zheng, **Feng-Chun Hsu**, Brandon Weissbourd, Kendyll Burnell, Elly Nedivi, Dushan N. Wadduwage, and Peter T. C. So. "High-Speed Computational Multiphoton Imaging Through Scattering Media." *Multiphoton Microscopy in the Biomedical Sciences XXIV*, Proc. SPIE 12847, PC1284709 (2024). *Presenter: Feng-Chun Hsu*
- Hao-Chung Chi, Anupama Nair, Yvonne Yuling Hu, **Feng-Chun Hsu**, Chia-Wei Hsu, Chun-Yu Lin, and Shean-Jen Chen. "Clear and Deep Temporal Focusing Multiphoton Microscopy Imaging Using Deep Prediction With PhyCell and ConvLSTM." *European Conference on Biomedical Optics*, 1263003 (2023).
- Wei-Shiuan Huang, Chia-Wei Hsu, **Feng-Chun Hsu**, Chun-Yu Lin, and Shean-Jen Chen. "Model Predictive Control-Based Adaptive Optics System With Deep-Learning Shack-Hartmann Wavefront Sensor." *Digital Optical Technologies 2023*, Proc. SPIE 12624, 126241C (2023).
- **Feng-Chun Hsu**, Chun-Yu Lin, Chia-Yuan Chang, and Shean-Jen Chen. "Shack-Hartmann-Based Wavefront and Intensity Sensing via U-Net." *Unconventional Optical Imaging III*, Proc. SPIE 12136, 121360A (2022). *Presenter: Feng-Chun Hsu*
- Liang-Wei Chen, **Feng-Chun Hsu**, Yvonne Yuling Hu, Chun-Yu Lin, and Shean-Jen Chen. "3D Micropatterned Multiphoton Stimulation via Deep Learning-Based Computer-Generated Holography With Temporal Focusing Confinement." *Biomedical Spectroscopy, Microscopy, and Imaging II*, Proc. SPIE 12144, 1214403 (2022).
- **Feng-Chun Hsu**, Yong Da Sie, Chun-Yu Lin, Yvonne Yuling Hu, and Shean-Jen Chen. "Light-Field Multiphoton Microscopy With Temporal Focusing-Based Volume-Selective Excitation." *Unconventional Optical Imaging III*, Proc. SPIE 12136, 121360F (2022). *Presenter: Feng-Chun Hsu*
- Liang-Wei Chen, **Feng-Chun Hsu**, Chun-Yu Lin, Yvonne Yuling Hu, and Shean-Jen Chen. "3D Micropatterned Multiphoton Stimulation via Deep Learning-Based Computer-Generated Holography With Temporal Focusing Confinement." *Optics & Photonics Taiwan International Conference* (2021).
- Sumesh Nair, Chia-Ying Chang, **Feng-Chun Hsu**, Ching-Chieh Su, and Shean-Jen Chen. "Rapid Laser Pest Control System With 3D Small-Object Detection." *AI and Optical Data Sciences*, Proc. SPIE 11299, 112990T (2020).
- **Feng-Chun Hsu**, Yong Da Sie, Chia-Yuan Chang, Chun-Yu Lin, Yvonne Y. Hu, and Shean-Jen Chen. "Advanced Volumetric Light-Field Microscopy With Temporal-Focusing Multiphoton Selective Excitation." *Three-Dimensional and Multidimensional Microscopy XXVII*, Proc. SPIE 11245, 112450G (2020). Conference presentation. *Presenter: Feng-Chun Hsu*
- Chia-Yuan Chang and **Feng-Chun Hsu**. "Hybrid Optoelectronic Signal Detection for Fluorescence Microscopy." *Optics & Photonics Taiwan International Conference* (2019).
- Xin-Ni Huang, Chun-Yu Lin, Yvonne Yuling Hu, **Feng-Chun Hsu**, and Shean-Jen Chen. "Improving Temporal-Focusing Multiphoton Microscopy Image Quality via Deep Learning." *Optics & Photonics Taiwan International Conference* (2019).
- **Feng-Chun Hsu**, Yong Da Sie, Feng-Jie Lai, and Shean-Jen Chen. "Volumetric Bioimaging Based on Light-Field Microscopy With Temporal-Focusing Illumination." *Three-Dimensional and Multidimensional Microscopy XXV*, Proc. SPIE 10499, 104991Q (2018). *Presenter: Feng-Chun Hsu*
- **Feng-Chun Hsu**, Yong Da Sie, Chia-Yuan Chang, Feng-Jie Lai, and Shean-Jen Chen. "Volumetric Bioimaging Based on Light-Field Microscopy With Temporal-Focusing Illumination." *Optics & Photonics Taiwan International Conference* (2017). *Presenter: Feng-Chun Hsu*

PATENTS

- Chun-Yu Lin, **Feng-Chun Hsu**, Sumesh Nair, Jin-Jie Su, Chia-Ying Chang, and Shean-Jen Chen. "Object Characteristic Locating Device and Laser and Imaging Integration System." US 11,127,160 B2, issued Sep 21, 2021.
- **Feng-Chun Hsu**, Chia-Yuan Chang, Jih-Liang Hsieh, Chun-Yu Lin, Chia-Ying Chang, and Shean-Jen Chen. "Laser and Imaging Integration System and Method Thereof." TW I727393 B, issued May 11, 2021.
- Chun-Yu Lin, **Feng-Chun Hsu**, Sumesh Nair, Jing-Jie Su, Chia-Ying Chang, and Shean-Jen Chen. "Object Characteristic Locating Device and Laser and Imaging Integration System." TW I706335 B, issued Oct 1, 2020.
- Shean-Jen Chen, Chia-Yuan Chang, **Feng-Chun Hsu**, and Yong Da Sie. "Position Inspection Method and Computer Program Product." TW I668411 B, issued Aug 11, 2019.